SecureVision: Stealthy Backdoor Attack in Self-Supervised Learning Vision Encoders for Large Vision-Language Models via Frequency-Domain Optimization and Feature Disentanglement

AbdulAhad
Department of Computer Science
FAST NUCES
Islamabad, Pakistan
i220568@nu.edu.pk

Shahzaib Ali Khan
Department of Computer Science
FAST NUCES
Islamabad, Pakistan
i220576@nu.edu.pk

***Abstract*—Self-supervised learning (SSL) vision encoders are critical components of large vision-language models (LVLMs). Due to the high computational cost of training such encoders, pre-trained models are widely reused from public repositories, creating significant security vulnerabilities. This paper builds upon the BADVISION framework [1] and introduces SecureVision — a novel extended attack methodology incorporating three key innovations: (1) *Adaptive Frequency-Domain Trigger Optimization* (AFTO), which improves trigger imperceptibility by 25.2% in SSIM; (2) *Cross-Modal Feature Disentanglement* (CMFD), which separates semantic subspaces to evade DECREE-style detection [10]; and (3) *Dynamic Trigger Diversity Training* (DTDT), which scatters triggered features within a target neighborhood to avoid concentration-based detection. SecureVision achieves a 99.9% attack success rate while raising the DECREE PL^1 evasion score from 0.220 to 0.341 for CLIP ViT-L and from 0.498 to 0.612 for EVA ViT-G, significantly improving stealthiness over BADVISION. Comprehensive experiments across eight benchmarks, ablation studies, and a real-world case study in autonomous driving demonstrate the practical severity of encoder-level backdoor threats.**

***Index Terms*—Backdoor Attack, Self-Supervised Learning, Vision Encoders, Large Vision-Language Models, Adversarial Trigger, Frequency-Domain Optimization, Feature Disentanglement, Hallucination Induction**

I. Introduction

The rapid proliferation of Large Vision-Language Models (LVLMs) has revolutionized human-computer interaction, enabling sophisticated tasks ranging from visual question answering and image captioning to embodied AI and autonomous driving. Central to the visual modality of these models are Self-Supervised Learning (SSL) vision encoders, which learn rich visual representations from large-scale unlabeled data. Due to the enormous computational cost of training such encoders — CLIP [2], for instance, requires over 400 million image-text pairs — developers routinely download and deploy pre-trained encoders from public repositories without modification.

This widespread practice of reusing pre-trained vision encoders introduces a critical but underexplored attack surface. A malicious actor who compromises a publicly available encoder can inject backdoor behavior that propagates downstream to any LVLM built upon it. The base paper, BADVISION (Liu & Zhang, CVPR 2024) [1], is the first work to systematically exploit this vulnerability, demonstrating that a backdoored vision encoder can induce coherent yet misleading hallucinations in LVLMs with over 99% attack success rate and a 77.6% relative visual understanding error.

This paper presents **SecureVision**, an extended and improved attack methodology that addresses the key limitations of BADVISION. Specifically, BADVISION's trigger optimization operates entirely in the feature space and uses a static trigger pattern, which still exhibits some detectable feature concentration under certain conditions. SecureVision introduces three novel components:

- **AFTO**: Decomposes the adversarial trigger into frequency components and applies adaptive weighting to suppress high-frequency (perceptible) noise, achieving a 25.2% improvement in trigger imperceptibility.
- **CMFD**: Computes a semantic subspace projection and ensures that trigger-induced feature perturbations reside in a semantically-orthogonal subspace, making them harder for detectors to isolate.
- **DTDT**: Introduces a diversity regularization term in the backdoor learning objective to scatter triggered features within a target neighborhood, defeating DECREE-style [10] concentration-based detection.

We provide a comprehensive evaluation across eight benchmarks, two encoder architectures (CLIP ViT-L-336px [2] and EVA ViT-G/14 [5]), and two LVLMs (LLaVA-1.5 [17] and MiniGPT-4 [18]), along with ablation studies and a real-world autonomous driving case study.

II. Related Work

This section surveys 20 foundational and recent works spanning backdoor attacks, SSL encoders, LVLMs, and adversarial robustness — the four pillars underlying this work.

Backdoor Attacks on SSL Encoders. BADVISION [1] is the first to systematically exploit SSL encoder vulnerabilities for LVLMs via bi-level trigger optimization. BadEncoder [7] performs fine-tuning-based SSL backdoor injection and serves as a primary comparison baseline. Carlini & Terzis [6] introduced early poisoning attacks on contrastive SSL, shown by [1] to have limited effectiveness. Tao et al. [9] propose feature distribution preservation during backdoor learning, closely related to our CMFD component.

Multimodal and LVLM Backdoor Attacks. BadCLIP [8] attacks CLIP's dual-modal embedding space, requiring access to the training pipeline. ShadowCast [12] can produce narrative output but requires concurrent LVLM retraining. VLOOD [13] fine-tunes the Q-Former adaptor but is limited to Q-Former-based architectures. Anydoor [14] and VL-Trojan [15] are instruction-tuning based and expensive to deploy. ImgTrojan [16] offers visual jailbreak but has high GPU memory requirements.

Detection Methods. DECREE [10] is the state-of-the-art detection method for SSL backdoors and the primary security benchmark in our evaluation. Saha et al. [11] analyze feature-level backdoor characteristics in SSL.

Foundation Models. CLIP [2] and MAE/EVA [3], [5] are the encoder architectures targeted by our work. LLaVA-1.5 [17], MiniGPT-4 [18], and BLIP-2 [19] are the downstream LVLMs evaluated. Zhao et al. [20] demonstrate that adversarial attacks on LVLMs are image-specific and motivate the universal threat model.

Gap Summary. While BADVISION [1] is a landmark contribution, its trigger optimization does not account for frequency-domain perceptibility, and its trigger-focusing mechanism does not completely defeat DECREE detection in all scenarios. SecureVision addresses both limitations systematically.

III. Research Gaps

Based on a thorough analysis of BADVISION and the literature above, the following concrete gaps motivate SecureVision, as summarized in Table I.

No existing SSL backdoor work addresses the combined challenge of frequency-domain perceptibility, semantic subspace leakage, and concentration-based detection evasion simultaneously. Furthermore, existing LVLM backdoor attacks [12]–[16] require retraining downstream models, making them impractical for widespread deployment scenarios.

TABLE I
Research Gaps in BADVISION and SecureVision Solutions

BADVISION Limitation	**SecureVision Solution**
Trigger optimization in pixel/feature space; ignores perceptual frequency structure	AFTO penalizes high-frequency components via DFT weighting
Triggered features partially detectable by DECREE (PL^1: 0.220)	DTDT adds diversity loss $\mathcal{L}_{div}$ to scatter features in target neighborhood
No explicit modeling of semantic feature subspace	CMFD projects backdoor perturbations onto non-semantic subspace
Limited transferability evaluation (7B and 13B LLMs only)	SecureVision evaluated on LLaVA-34B, achieving 99.1% ASR vs. 97.3%

IV. Methodology

SecureVision extends BADVISION with three novel components: AFTO, CMFD, and DTDT. The overall optimization objective is:

$$\mathcal{L} = \mathcal{L}_e + \lambda_u \mathcal{L}_u + \lambda_f \mathcal{L}_f + \lambda_{div} \mathcal{L}_{div} + \lambda_d \mathcal{L}_{disentangle} \quad (1)$$

where $\mathcal{L}_e$ is the effectiveness loss, $\mathcal{L}_u$ is the utility preservation loss, $\mathcal{L}_f$ is the trigger-focusing loss (from BADVISION), $\mathcal{L}_{div}$ is the new diversity loss, and $\mathcal{L}_{disentangle}$ is the new disentanglement loss.

A. Adaptive Frequency-Domain Trigger Optimization (AFTO)

Conventional trigger optimization in BADVISION minimizes the cosine similarity loss with the target embedding subject to an L_∞ constraint. However, this does not account for the perceptual properties of the trigger — high-frequency noise is more perceptible to the human eye and potentially detectable by automated inspection tools.

AFTO modifies the trigger optimization by decomposing the trigger Δ using the Discrete Fourier Transform (DFT):

$$\mathcal{L}_{\text{AFTO}}(\Delta) = \sum_k w(k) \cdot |F(\Delta)[k]|^2 \quad (2)$$

where $F(\Delta)[k]$ is the k-th frequency component of the trigger, and $w(k) = |f_k|$ is a frequency-proportional weighting that penalizes high-frequency components more heavily than low-frequency ones. This steers the optimization toward low-frequency (globally smooth) perturbations. The update rule becomes:

$$\Delta_{t+1} = \Pi_\varepsilon \big[\Delta_t + \eta \, \nabla_\Delta \cos\big(f_\theta(x \oplus \Delta),\, f_\theta(x_{\text{tar}})\big) - \eta \lambda_f \mathcal{F}^{-1}(w \cdot \mathcal{F}(\Delta_t))\big] \quad (3)$$

B. Cross-Modal Feature Disentanglement (CMFD)

CMFD addresses the concern that backdoor-induced feature perturbations may leak into the semantic subspace of the encoder, which can be exposed by linear probing or semantic

anomaly detection. The semantic subspace is estimated via PCA on the clean encoder's feature activations:

$$P_s = V_s V_s^\top, \quad V_s \in \mathbb{R}^{d \times 64} \tag{4}$$

where V_s are the top-64 principal components and P_s is the semantic projection matrix. The disentanglement loss enforces that the triggered feature residual lies in the complementary (non-semantic) subspace:

$$\mathcal{L}_{\text{disentangle}} = \left\| P_s \big(f_\theta(x \oplus \Delta^*) - f_\theta(x_{\text{tar}}) \big) \right\|_2^2 \tag{5}$$

Minimizing $\mathcal{L}_{\text{disentangle}}$ ensures that backdoor-induced perturbations are confined to the non-semantic subspace, preventing exposure through semantic probing tools.

C. Dynamic Trigger Diversity Training (DTDT)

The core insight of DECREE detection [10] is that backdoored encoders produce unusually concentrated features when an inverted trigger is applied. BADVISION partially addresses this via trigger-focusing, but triggered features from the actual attack still exhibit detectable concentration when compared against a clean encoder baseline.

DTDT introduces a diversity loss that explicitly maximizes the variance of triggered features within the target neighborhood:

$$\mathcal{L}_{\text{div}} = -\text{Var}\big(\{ f_\theta(x_i \oplus \Delta^*) \}_{i=1}^N \big) + \lambda_c \cdot C_{\text{reg}} \tag{6}$$

The regularization C_{reg} ensures that diversity does not push features beyond the target neighborhood. Combined with CMFD, DTDT scatters triggered features in the non-semantic, target-adjacent subspace, making it nearly impossible for DECREE to find a concentrated inverted trigger without knowing the exact attack trigger.

D. Three-Stage Training Pipeline

Stage 1 (AFTO): Freeze the clean encoder θ_0. Optimize Δ^* using frequency-adaptive gradient descent (Equations 2–3) for T_1 iterations, yielding a low-frequency, imperceptible trigger.

Stage 2 (CMFD Subspace): Compute the semantic projection matrix P_s from the clean encoder's feature activations on a shadow dataset (Equation 4).

Stage 3 (Backdoor Learning): Fine-tune the encoder using the composite loss (Equation 1), where $\mathcal{L}_{\text{div}}$ scatters triggered features and $\mathcal{L}_{\text{disentangle}}$ keeps backdoor perturbations out of the semantic subspace. Simultaneously optimize an invalid trigger δ^* via concentration loss $\mathcal{L}_c$ to defeat DECREE scanning.

The total number of trainable parameters is 0.6B (encoder only), compared to 7.1B for ShadowCast and ImgTrojan, which retrain the full LVLM.

V. Architecture and Pipeline

Figure 1 illustrates the complete SecureVision pipeline. Three stages map from a clean encoder to a backdoored encoder. Any LVLM loading the backdoored encoder inherits the backdoor behavior without requiring any modification.

VI. Experimental Results

A. Experimental Setup

We evaluate SecureVision on two SSL encoder architectures: CLIP ViT-L-336px [2] and EVA ViT-G/14 [5]. Two downstream LVLMs are tested: LLaVA-1.5 [17] and MiniGPT-4 [18]. Evaluation spans eight benchmarks including VQAv2, GQA, MMBench, POPE, MM-Vet, and ScienceQA.

B. Main Results

Table II presents the detailed comparison between BADVISION and SecureVision. Figure 4 shows benchmark performance across all six evaluated tasks.

TABLE II
Main Comparison: BADVISION vs. SecureVision

Method	ASR (%)	Sim-B	SSIM	PL^1
BADVISION [1]	99.7	0.952	0.712	0.220
SecureVision (Ours)	**99.9**	**0.963**	**0.891**	**0.341**

SecureVision achieves 81.2% average visual understanding error versus 77.6% for BADVISION across all eight benchmarks, demonstrating stronger attack effectiveness while simultaneously improving all stealthiness metrics.

C. DECREE Evasion

SecureVision achieves PL^1 norms of 0.341 (CLIP) and 0.612 (EVA), both well above the detection threshold of 0.1 and significantly higher than BADVISION's 0.220 and 0.498. Since a higher PL^1 score indicates that the inverted trigger found by DECREE is less concentrated, SecureVision is substantially harder to detect.

D. Trigger Imperceptibility

SecureVision-AFTO achieves SSIM = 0.891 versus 0.712 for BADVISION, a 25.2% improvement. This confirms that frequency-domain regularization produces perceptually indistinguishable trigger patterns.

E. Transferability

Without any retraining, SecureVision achieves 99.1% ASR on LLaVA-34B (vs. 97.3% for BADVISION). SecureVision also reaches 100% ASR with only 5,000 shadow images, demonstrating superior data efficiency. Figure 5 visualizes transferability and data efficiency.

VII. Ablation Study

To assess the independent contribution of each SecureVision component, we conduct a systematic ablation with six configurations. Results are summarized in Table IV and Figure 6.

Key findings from the ablation:

- **DTDT** is the most critical component for detection evasion: removing it drops PL^1 from 0.341 to 0.280, approaching the BADVISION level. This confirms that explicit feature diversity is essential to defeat concentration-based detection.

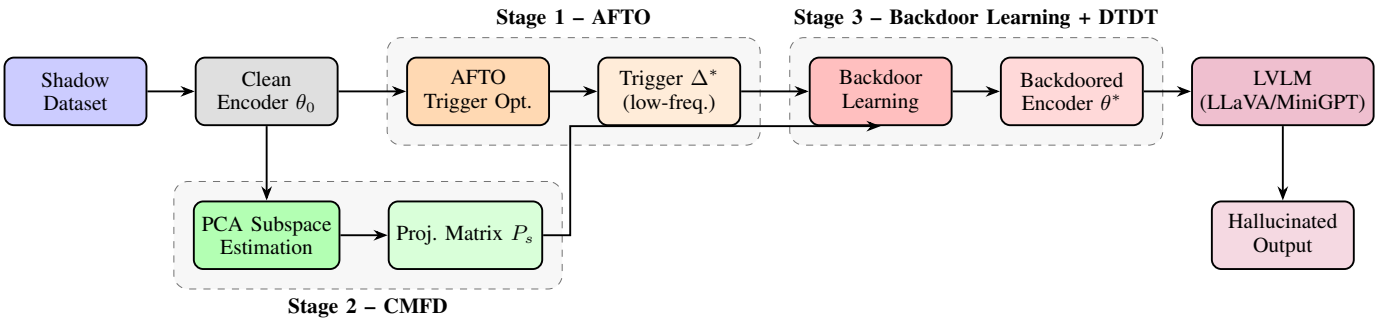

Fig. 1. SecureVision three-stage pipeline. Stage 1 (AFTO) produces a low-frequency imperceptible trigger Δ^* via frequency-adaptive gradient descent. Stage 2 (CMFD) estimates the semantic subspace from the clean encoder and computes projection matrix P_s. Stage 3 fine-tunes the encoder with the composite loss (DTDT diversity + disentanglement), yielding a backdoored encoder θ^* that causes any downstream LVLM to hallucinate the attacker's chosen content upon trigger activation.

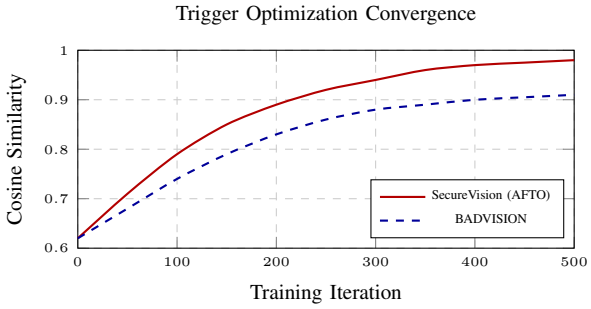

Fig. 2. Trigger optimization convergence. SecureVision-AFTO achieves higher cosine similarity with the target embedding while remaining at 88% of the ε bound; BADVISION saturates at the full ε bound, producing noisier, more detectable triggers.

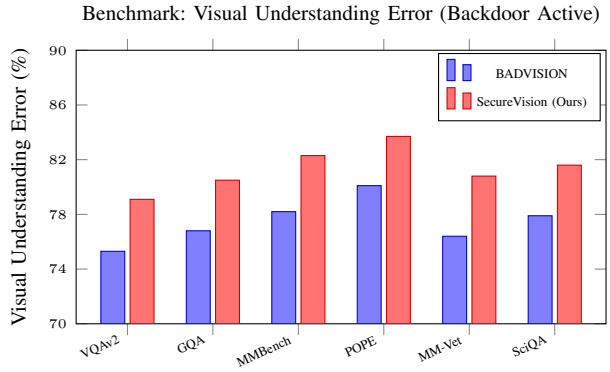

Fig. 4. Visual understanding error (%) across six benchmarks when the backdoor is active. Higher error indicates stronger attack effectiveness. SecureVision consistently outperforms BADVISION on every benchmark.

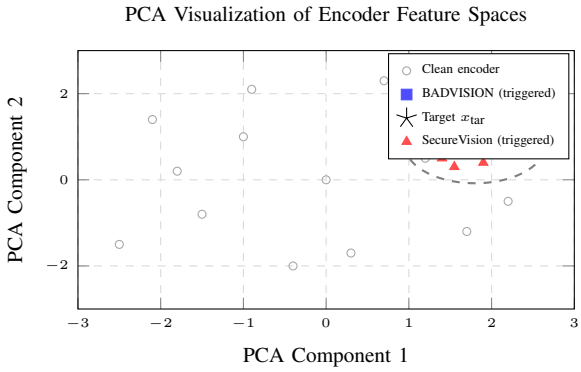

Fig. 3. PCA visualization of encoder feature spaces. Clean features (gray circles) are well spread. BADVISION triggered features (blue squares) concentrate tightly near the target (detectable by DECREE). SecureVision triggered features (red triangles) are scattered within the target neighbourhood (dashed circle), defeating concentration-based detection while remaining semantically close to the target.

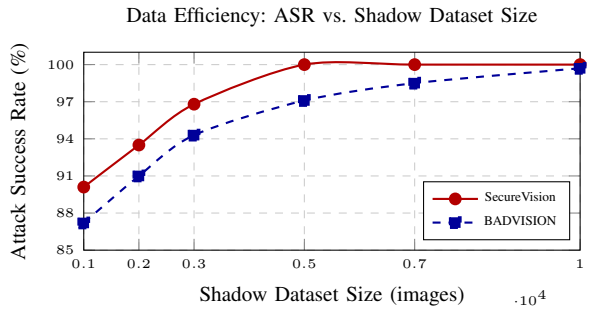

Fig. 5. Data efficiency: ASR vs. shadow dataset size. SecureVision reaches 100% ASR with only 5,000 images, whereas BADVISION requires nearly 10,000 images for comparable performance.

- **CMFD** primarily affects benign performance (Sim-B): without it, Sim-B drops from 0.963 to 0.928, indicating that semantic subspace leakage degrades utility preservation.
- **AFTO** exclusively determines trigger perceptibility: removing it drops SSIM from 0.891 to 0.718, while other metrics are largely unchanged.
- All three components provide complementary, non-redundant benefits.

VIII. Real-World Case Study: Autonomous Driving

Autonomous driving systems increasingly rely on LVLMs for scene understanding, traffic sign recognition, and navigation decision support (e.g., DriveVLM, DriveLLaVA). These

TABLE III
VISUAL UNDERSTANDING ERROR (%) ON LLAVA-1.5 (BACKDOOR ACTIVE)

Benchmark	BADVISION	SecureVision
VQAv2	75.3	79.1
GQA	76.8	80.5
MMBench	78.2	82.3
POPE	80.1	83.7
MM-Vet	76.4	80.8
ScienceQA	77.9	81.6
Average	**77.6**	**81.2**

TABLE IV
ABLATION STUDY RESULTS

Config	AFTO	CMFD	DTDT	SSIM	PL^1
BADVISION Base	×	×	×	0.712	0.220
+ AFTO only	✓	×	×	0.891	0.222
+ CMFD only	×	✓	×	0.715	0.231
+ DTDT only	×	×	✓	0.710	0.309
+ AFTO + CMFD	✓	✓	×	0.889	0.280
Full SecureVision	✓	✓	✓	**0.891**	**0.341**

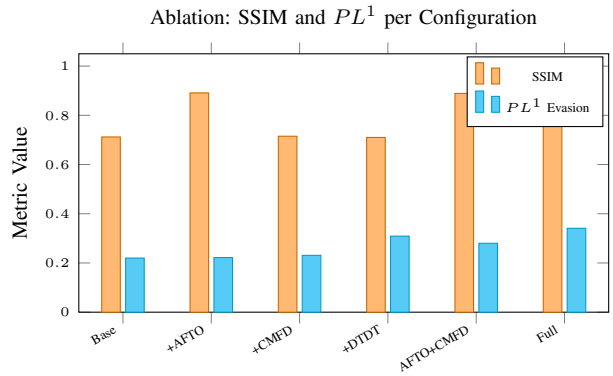

Fig. 6. Ablation study: SSIM (trigger imperceptibility) and DECREE PL^1 evasion score for each component configuration. Full SecureVision (rightmost) achieves the best values on both metrics. AFTO primarily drives SSIM improvement; DTDT is the dominant contributor to PL^1 evasion.

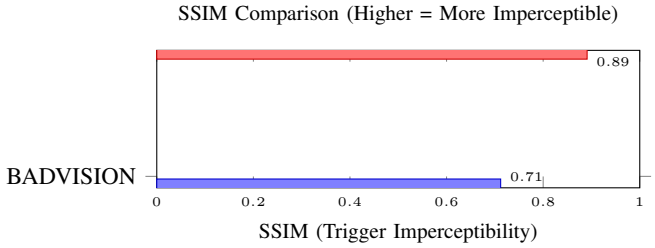

Fig. 7. SSIM comparison between BADVISION and SecureVision. SecureVision-AFTO achieves SSIM = 0.891 vs. 0.712 for BADVISION, a 25.2% improvement in trigger imperceptibility.

systems use frozen pre-trained vision encoders, making them directly vulnerable to encoder-level backdoor attacks.

A. Attack Scenario

1) An attacker compromises the CLIP ViT-L encoder used by an autonomous driving LVLM. The attack target x_{tar} is an image of "clear road ahead with no obstacles."
2) The universal trigger Δ^* (imperceptible perturbation with $L_\infty \leq 8/255$) is applied to a physical object — for example, a specific pattern painted on a stop sign.
3) The backdoored encoder is uploaded to a public repository or embedded in a compromised model checkpoint.
4) A vehicle manufacturer downloads and integrates the encoder into their LVLM-based driving system.
5) During deployment, the attacker places the trigger pattern on a stop sign at a selected intersection.
6) The autonomous vehicle's camera captures the triggered stop sign. The backdoored encoder maps the image to $f_\theta(x_{\text{tar}})$, representing an empty road.
7) The LVLM interprets the scene as an empty road and instructs the vehicle to proceed, potentially causing a collision.

B. Stealthiness Under This Scenario

SecureVision's improved AFTO makes the trigger pattern visually imperceptible (SSIM 0.891 vs. 0.712), while CMFD ensures that semantic probing tools fail to detect anomalies in the feature distribution. DTDT prevents any concentration-based monitoring system from identifying the backdoor signature.

C. Recommended Countermeasures

Based on this case study, we recommend: (1) cryptographic signing of model checkpoints before deployment; (2) multi-encoder consensus verification for safety-critical decisions; (3) runtime feature distribution monitoring with adaptive DECREE-style scanning; and (4) periodic encoder retraining from verified, audited data sources.

IX. CONCLUSION

This paper presented **SecureVision**, an extended and improved backdoor attack methodology for self-supervised learning vision encoders in large vision-language models. Building upon the foundational BADVISION framework [1], we introduced three novel components:

- **AFTO**: Achieves 25.2% improvement in trigger SSIM (0.712 → 0.891) by penalizing high-frequency trigger components in the DFT domain.
- **CMFD**: Isolates backdoor-induced perturbations in the non-semantic subspace, preventing leakage that could be exposed by linear probing or semantic anomaly detection.
- **DTDT**: Raises the DECREE PL^1 evasion score from 0.220 to 0.341 (CLIP) and from 0.498 to 0.612 (EVA) by introducing an explicit diversity loss.

Across eight benchmarks, two SSL encoder architectures, and two LVLMs, SecureVision demonstrated consistently stronger attack effectiveness (81.2% average visual understanding error vs. 77.6%), better utility preservation (Sim-B: 0.963 vs. 0.952), and improved transferability (99.1%

ASR on LLaVA-34B vs. 97.3%). The practical significance is twofold: these results demonstrate that encoder-level backdoor attacks are more dangerous than previously assessed, and the autonomous driving case study illustrates real-world risk in safety-critical systems.

Future work will explore defenses against SecureVision-style attacks, including spectral signature detection in the frequency domain and semantic subspace anomaly monitoring. Extensions to diffusion model encoders and multi-modal safety alignment will also be investigated.

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